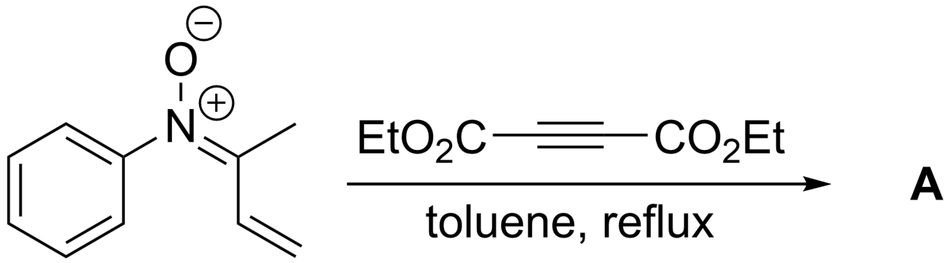
Question

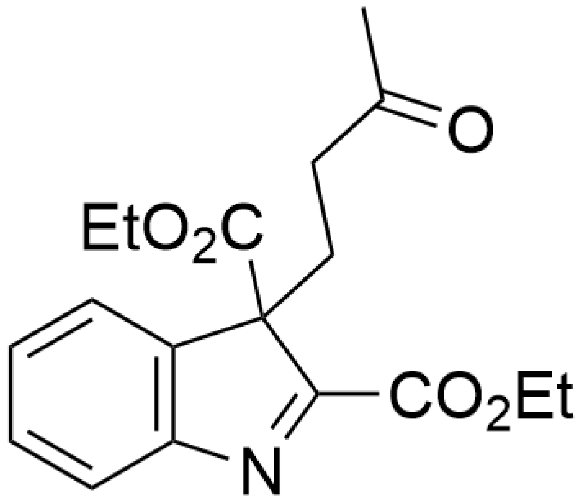


[O-]/[N+](C1=CC=CC=C1)=C(C)/C=C>O=C(C#CC(OCC)=O)OCC,toluene,reflux>[**A**],**A** is the reaction product

It is believed that the substrate reacts with diethyl butynedioate to generate a seven-membered ring intermediate **X**; **X** is converted to the final product under the action of trace water in the system. It is known that the reaction product **A** contains 2 rings, and the molecular formula of **A** is $C_{18}H_{21}NO_5$, try to give the structural formula of **A**.

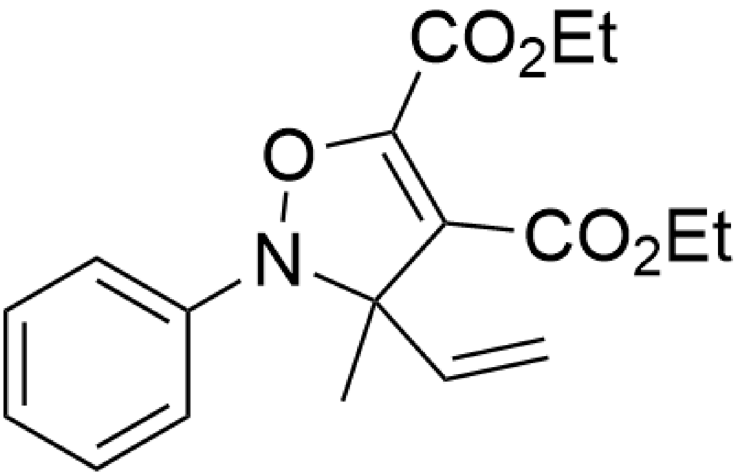
A. All other options are incorrect

B.



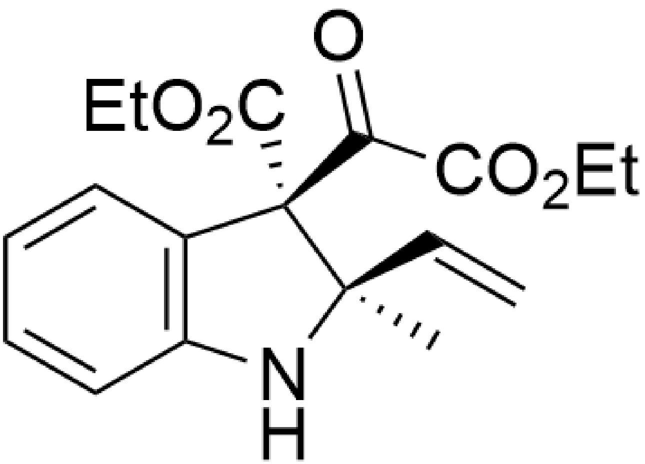
O=C(C)CCC1(C(OCC)=O)C2=C(N=C1C(OCC)=O)C=CC=C2

C.



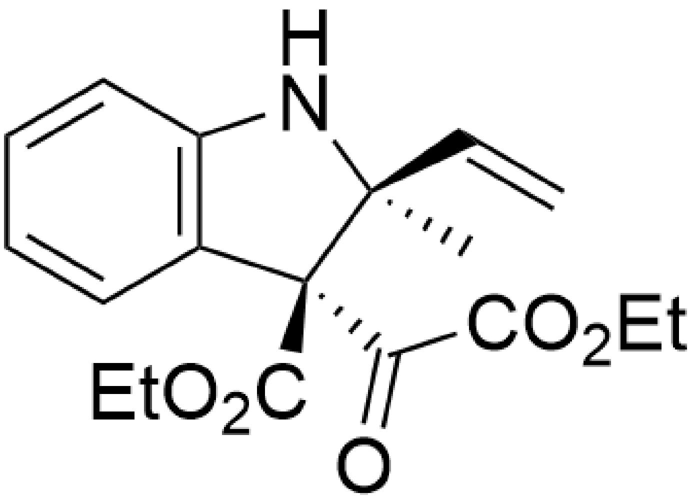
CC1(C=C)N(OC(C(OCC)=O)=C1C(OCC)=O)C2=CC=CC=C2

D.



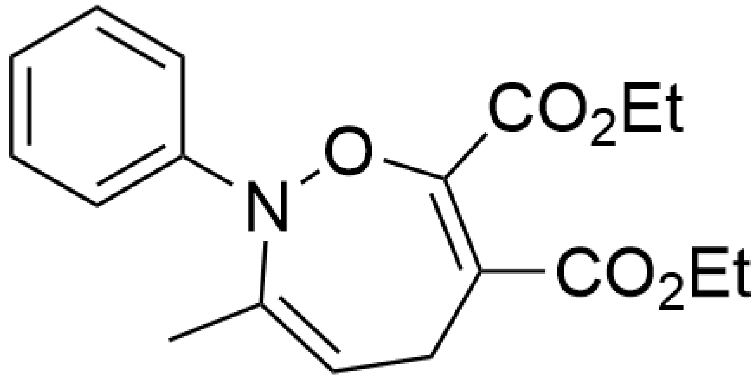
O=C(C(OCC)=O)[C@@]1(C(OCC)=O)[C@](C=C)(C)NC2=C1C=CC=C2

E.



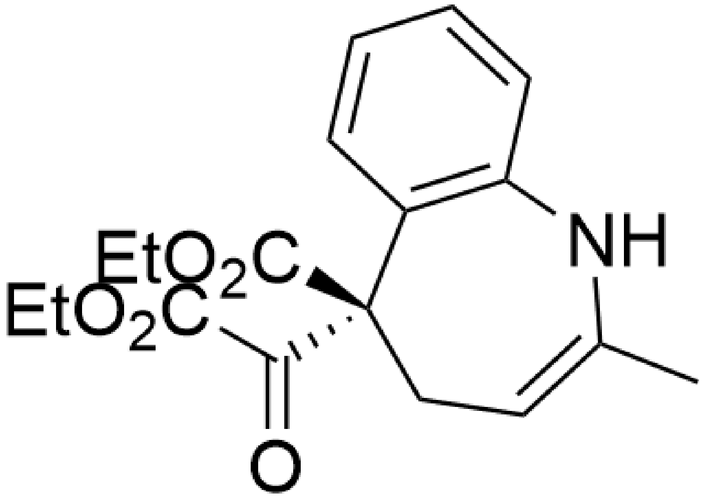
O=C(C(OCC)=O)[C@@]1(C(OCC)=O)[C@@](C=C)(C)NC2=C1C=CC=C2

F.



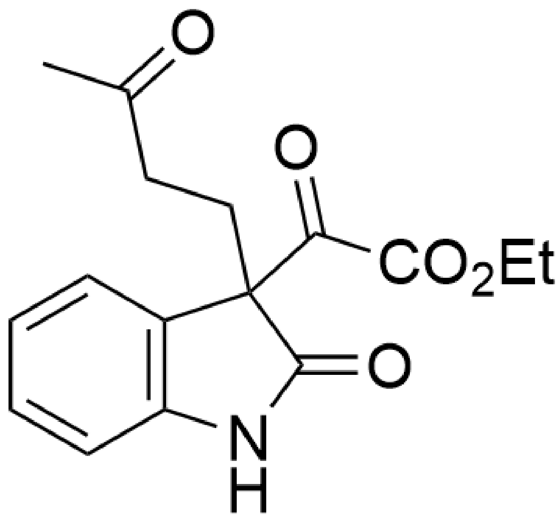
CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

G.



O=C([C@@]1(C2=C(NC(C)=CC1)C=CC=C2)C(OCC)=O)C(OCC)=O

H.



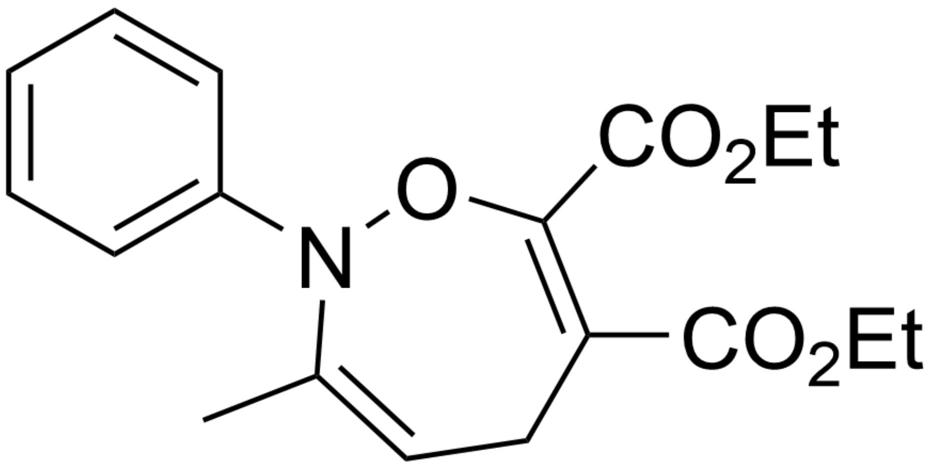
O=C(C(OCC)=O)C1(CCC(C)=O)C2=C(NC1=O)C=CC=C2

Answer

Correct Answer: B

Detailed Explanation

First, according to the problem prompt, the reaction substrate reacts with diethyl butynedioate to generate the seven-membered ring intermediate **X**

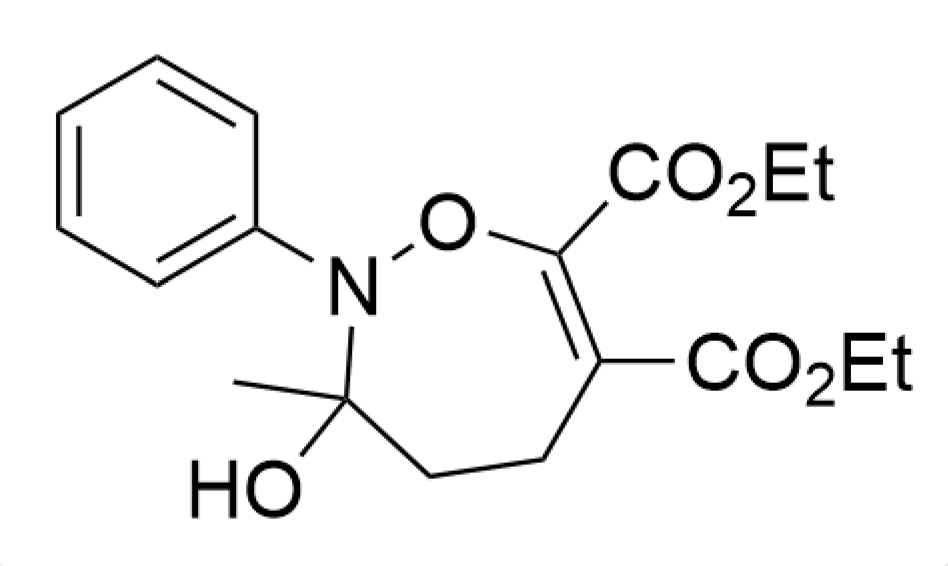


Intermediate **X**: CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

CHECKPOINT 1 PTS

Intermediate **X** : CC1=CCC(C(OCC)=O)=C(C(OCC)=O)ON1C2=CC=CC=C2

Then, intermediate **X** first forms intermediate **1** under the action of trace water.

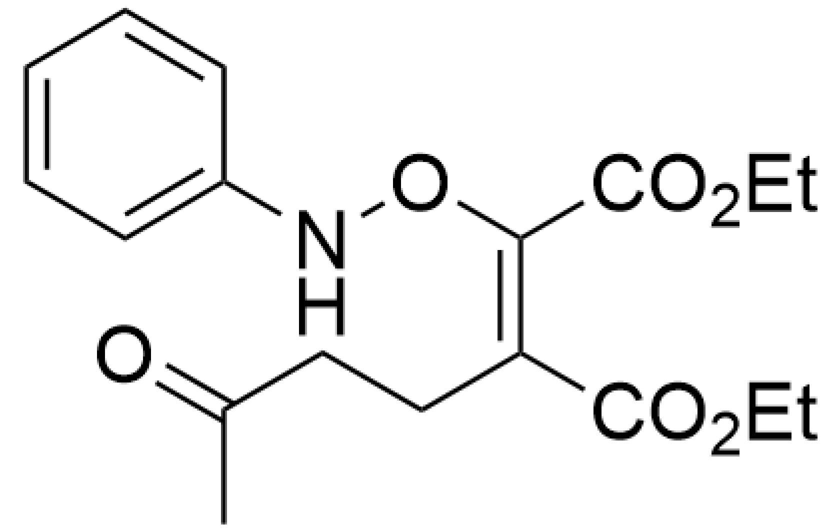


Intermediate **1**: CC1(O)N(OC(C(OCC)=O)=C(CC1)C(OCC)=O)C2=CC=CC=C2

CHECKPOINT 1 PTS

Intermediate **1**: CC1(O)N(OC(C(OCC)=O)=C(CC1)C(OCC)=O)C2=CC=CC=C2

Then, intermediate **1** further undergoes ring-opening to obtain intermediate **2**

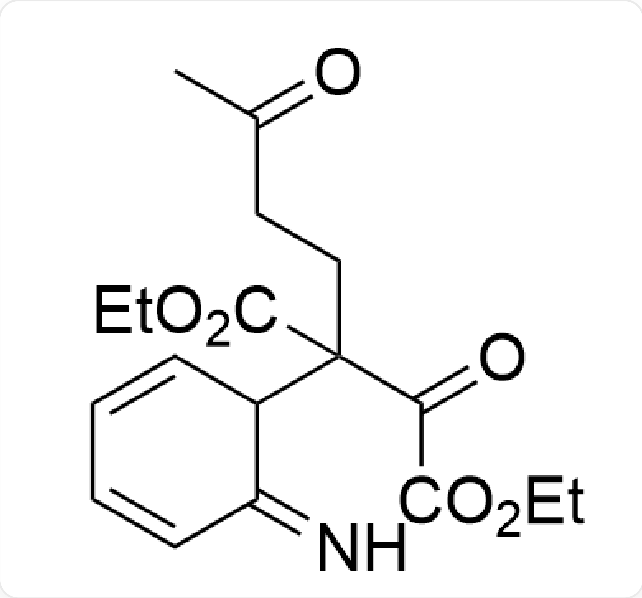


Intermediate **2**: CC(CC/C(C(OCC)=O)=C(C(OCC)=O)\ONC1=CC=CC=C1)=O

CHECKPOINT 1 PTS

Intermediate **2**: CC(CC/C(C(OCC)=O)=C(C(OCC)=O)\ONC1=CC=CC=C1)=O

At this time, the conformation of intermediate **2** allows it to undergo a 3,3-σ rearrangement reaction to obtain intermediate **3**

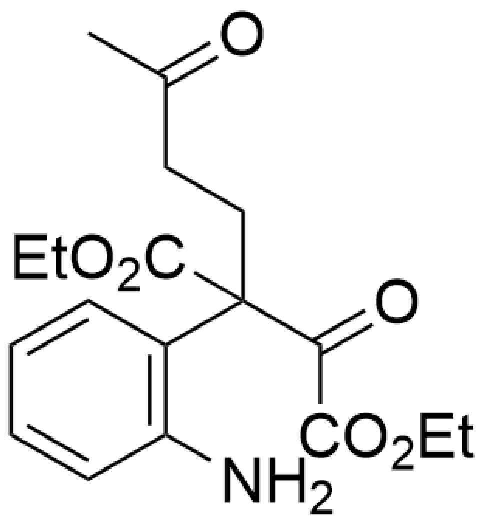


Intermediate $\mathbf{3}$: N=C1C=CC=CC1C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O

CHECKPOINT 1 PTS

Intermediate $\mathbf{3}$: N=C1C=CC=CC1C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O

Intermediate $\mathbf{3}$ undergoes aromatization to obtain intermediate $\mathbf{4}$

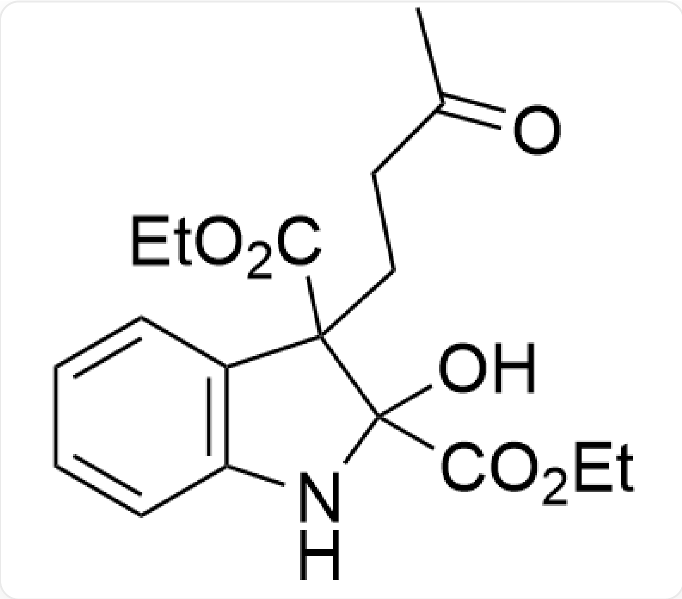


Intermediate **4**：NC1=C(C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O)C=CC=C1

CHECKPOINT 1 PTS

Intermediate **4**： NC1=C(C(CCC(C)=O)(C(OCC)=O)C(C(OCC)=O)=O)C=CC=C1

The carbonyl group in intermediate **4** has strong reactivity, so the amino group attacks the carbonyl group to form a five-membered ring, yielding intermediate **5**

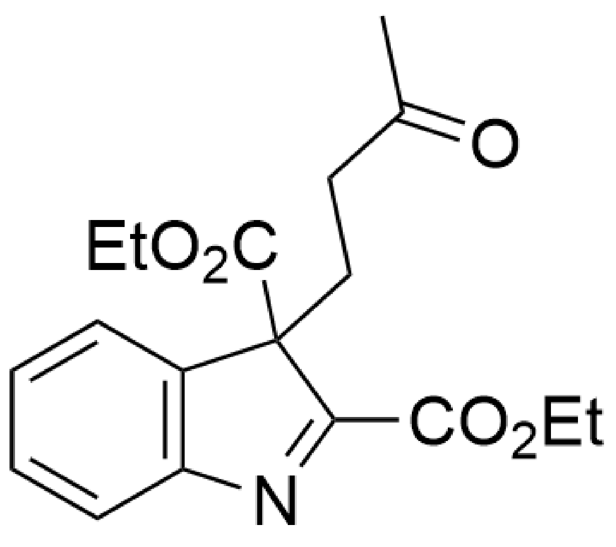


Intermediate $\mathbf{5}$: OC1(C(OCC)=O)C(CCC(C)=O)(C(OCC)=O)C2=C(N1)C=CC=C2

CHECKPOINT 1 PTS

Intermediate $\mathbf{5}$: OC1(C(OCC)=O)C(CCC(C)=O)(C(OCC)=O)C2=C(N1)C=CC=C2

Intermediate $\mathbf{5}$ continues to lose a molecule of water to obtain the final product $\mathbf{A}$



Product **A**: O=C(C)CCC1(C(OCC)=O)C2=C(N=C1C(OCC)=O)C=CC=C2